PTERO Desktop

Practical user tutorial | v0.2.0 | Windows private beta

For colleagues testing Bruker .d and Thermo .raw datasets

PTERO — Proteomics Tool for Exploratory Research & Optimization — brings raw-data exploration, identification review and differential protein abundance into a desktop interface. This guide explains what to try, what the displays mean and what feedback will help improve the software.

1 Install and launch

- Extract the entire supplied PTERO_v1_rc_windows_x64.zip archive into a normal folder. Do not launch the executable from inside the ZIP.
- Open the extracted PTERO_v1_rc folder and double-click PTERO_v1_rc.exe. The older executable filename is normal for this validated v0.2.0 beta.
- Keep the extracted folder together; the executable relies on the accompanying files. Initial startup may take some time.
- Use a local folder with enough free disk space for the converted dataset. Record your laptop RAM, CPU and available storage for feedback.

2 Prepare your instrument data

Bruker timsTOF — .d folder

In Raw data → Create PTERO Dataset, choose the complete Bruker .d directory, set a new output .zarr folder and select the MS level appropriate to your question (MS1, MS2, or both if offered). Start conversion and wait for completion before opening the output in Open PTERO Dataset. For DDA/DIA runs, the available precursor or window-scheme display depends on the acquisition mode.

Thermo — .raw file

The current PTERO converter does not accept Thermo .raw directly. Convert it externally to .mzML using a compatible converter such as ProteoWizard msConvert, with the correct vendor reader installed where required. Check the resulting mzML before proceeding. Then use Create PTERO Dataset with the mzML input, create a new .zarr folder, and open that folder. Preserve the original .raw and mzML files.

Tip: first try one representative run. Conversion time and output size are part of the evaluation; do not assume a conversion is stuck just because a large file takes time.

3 Raw data: what every view does

Chromatogram + spectrum

Browse the chromatographic signal; click the chromatogram to inspect the corresponding scan spectrum. Try a peak you recognise.

Feature map (RT × m/z)

Visualise intensity distribution across retention time and m/z; look for interpretable patterns.

Mobility (m/z × 1/K0)

When ion mobility is present, inspect how signals distribute across m/z and inverse mobility.

RT × 1/K0

Inspect the retention-time / ion-mobility footprint when supported by the dataset.

XIC

Enter one target m/z to extract a chromatogram; with mobility data, inspect mobilogram/footprint. Enter comma-separated m/z targets to compare traces. Narrow RT first for large runs.

Diagnostics

Review acquisition characteristics including peaks per scan and MS1 cycle time.

Precursor (DDA)

Review MS2 precursor m/z distributions and their relationship to retention time.

Window scheme (DIA)

Inspect DIA isolation windows; this describes the acquisition scheme rather than a single selected RT window.

QC

Read acquisition quality metrics. Some counts/cycle metrics cover the whole run; peak-derived values may follow the current MS level and range.

Sample comparison

Open two or more PTERO Datasets and compute identification-free QC metrics, variability and potential outlier flags. Flags invite inspection, not automatic rejection.

4 Identifications

Use the Identifications main page to load supported identification results. The interface offers a PTERO Dataset source selector, paths for identification text files and a FASTA sequence file, and a Load identifications action. Provide the files that correspond to the run you are exploring.

- Use the run and protein selectors and the protein search to locate a known identification.
- Select a protein to inspect its peptides and sequence coverage.
- Select a peptide to examine its annotated MS/MS spectrum and, where available, its precursor extracted-ion chromatogram.
- Check whether spectrum matching and annotation make sense for your own identification files; report mismatches with reproducible steps.

5 Differential Protein Abundance

Open Differential Protein Abundance and load a supported protein-quantity table. Review the quantity selection and the automatically suggested sample groups; correct the design and choose the reference (baseline) group before analysis.

- Choose the analysis method and track available in your build; interpret missing-value handling and imputation cautiously.

- Use the adjustable adjusted-p threshold, absolute log2 fold-change threshold, on/off minimum-presence control and heatmap size to explore results.
- Inspect the volcano plot, on/off detection view, top-protein heatmap, results table and per-protein detail/boxplot.
- Export results and graphics if needed. A visual signal is not proof of biological validity: check the experimental design and assumptions.

6 What to test on a normal laptop

- Start-up: How long until the application is usable? Any Windows security or dependency problems?
- Conversion: Input type and size, conversion time, output .zarr size, peak RAM if available, and any errors.
- Exploration: Time to open a converted dataset; responsiveness when changing RT/MS filters, clicking spectra and extracting XICs.
- Scientific usefulness: Which plots, QC measures and identification/abundance views are genuinely useful? What is missing?
- Hardware: Windows version, CPU, RAM, free disk, and whether the application remained usable alongside normal laptop tasks.

7 Send feedback

Five-minute report: (1) data type and approximate size; (2) laptop specifications; (3) what worked and how fast it felt; (4) confusing or missing functions; (5) one improvement you would prioritise. Include screenshots and steps to reproduce errors. Avoid sharing confidential raw data without permission.

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